TomTom Hazard Warnings

– Technical specification –
– B2B Delivery –

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Document history

Version	Date	Changes
1.0	2023-04-19	• Initial version of the document
1.1	2023-06-05	• Emergency Vehicle Approaching added
1.2	2023-10-24	• Wrong Way Driver added
1.3	2024-06-19	• OSM Way ID references added • Start time statement changed • Adaption to the new topic naming convention

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2 Scope

This document describes the technical specification of the TomTom Hazard Warnings service offered by TomTom.

All examples provided in this document are informative and may not be complete.

3 Introduction to the TomTom Hazard Warnings service

The TomTom Hazard Warnings service is a low-latency warning service, mainly for end-user devices. The warnings published by the service are safety related and request the end-user to pay special attention to.

The hazard warnings service helps to anticipate the road ahead. Based on probe data and enriched with government feeds, community input, and car sensor data the warnings are meant to be on display or to an automated driving system to adjust speed or adapt functionality.

TomTom Hazard Warnings service provides alerts on safety-critical incidents in the following categories:

- Traffic hazards
- Road hazards
- Weather hazards
- Generic hazards

3.1 Hazard types

The TomTom Hazard Warnings service offers different types of hazards.

Hazards type	Category	Description
Accident	Traffic	Accidents are dangerous situations, especially if the accident happened very recently. Situations like this are usually unclear; other drivers need to be very careful to pass the location or offer assistance.
Broken-down vehicle	Traffic	Vehicles that come to a sudden stop may cause an obstacle. Furthermore, the passengers of the broken-down vehicle may leave the vehicle and other drivers need to take special care.
Emergency Vehicle Approaching	Traffic	Emergency Vehicle Approaching (EVA) indicates vehicles that have priority on the road. Drivers need to let them pass.
Jam tails	Traffic	Jam tails are at the end of queues, which can be dangerous. Drivers are warned to reduce speed ahead of the vehicles which are already queuing in a jam.



Wrong Way Driver	Traffic	Vehicles driving in the wrong way, against the legal traffic flow of the road.
Slippery road	Weather	Slippery road conditions are mainly caused by bad weather conditions, and various substances, such as oil or sand on the road surface. Such conditions can lead to unexpected movements of the vehicle.
Reduced visibility	Weather	Severe weather conditions may reduce the visibility to an extent where drivers must be careful and should reduce speed and watch out for other vehicles.
Strong winds	Weather	Unexpected strong winds (e.g., cross winds) may cause vehicles to move in undesired directions. Drivers should be aware and reduce speed as necessary.
Bad road conditions	Road	The road surface may be damaged by potholes or large cracks, causing drivers to reduce speed and be alert.
Objects on road	Road	Any kind of objects on a road (like animals, boxes, or people create dangerous situations for drivers. Drivers need to be alert and carefully pass road obstacles.
Roadworks	Road	Roadworks are construction sites for roadway maintenance. They include reduced speed and often have lane changes.
Generic	Generic	All kind of other hazards which cannot be classified with the previous descriptions.

Table 1: Hazard types

The list of hazard types will over the coming months. TomTom is currently looking into more types and warnings and will inform its customers about new types before its launch.

4 Service API

TomTom Hazard Warnings are delivered via the TomTom B2B delivery platform. To connect to the platform please consult the documentation of the platform.

After successful connection to the TomTom B2B delivery platform Hazard Warning events are available by subscription to the topics with the format:

<data_channel>.tomtom.hazards.<content_api_version>.<product_configuration>.<region>.<hazard_type>

For Hazards, the following values are defined:

Field	Description
data_channel	channel or environment of the data
content_api_version	an identifier of the schema version of the content, breaking changes will be reflected in changed Content API versions, current value “v1”
product_configuration	the type of the product customized using a various set of feature toggles
region	the identifier of a regional partition of the content, can be ISO 2 country code or city name, e.g. “JP”, will be shared with customers for individual access
hazard_type	The type of hazard like SLIPPERY_ROAD

Hazard Type: Hazard type identifier, based on list of Table 1: Hazard types.

Example of all topics for an evaluation channel in the region “JP”:

channel_evaluation.tomtom.hazards.v1.all.JP.SLIPPERY_ROAD
channel_evaluation.tomtom.hazards.v1.all.JP.GENERIC
channel_evaluation.tomtom.hazards.v1.all.JP.BROKEN_DOWN_VEHICLE
channel_evaluation.tomtom.hazards.v1.all. JP.EMERGENCY_VEHICLE_APPROACHING
channel_evaluation.tomtom.hazards.v1.all. JP.WRONG_WAY_DRIVER
channel_evaluation.tomtom.hazards.v1.all. JP.BAD_ROAD_CONDITIONS
channel_evaluation.tomtom.hazards.v1.all. JP.ACCIDENT
channel_evaluation.tomtom.hazards.v1.all. JP.ROADWORKS
channel_evaluation.tomtom.hazards.v1.all. JP.OBJECTS_ON_ROAD
channel_evaluation.tomtom.hazards.v1.all. JP.REDUCED_VISIBILITY
channel_evaluation.tomtom.hazards.v1.all. JP.JAM_TAIL_WARNING
channel_evaluation.tomtom.hazards.v1.all. JP.STRONG_WIND

5 Data Format

Data from the delivery platform will be delivered encoded in Protocol Buffers format version 3. The content is structurally described in the form of Protocol Buffers schema files. The Schemas are provided by TomTom.

5.1 HazardsReport

Main component of the Hazards Protocol Buffers schema. It encapsulates Metadata component and HazardMessage component.

Attribute	Data type	Description
metaData	MetaData	See 5.2
message	HazardMessage	See 5.3

5.2 MetaData

Metadata component contains metadata about the Hazard Message. It contains creation and expiration times of the message.

Attribute	Data type	Description
creationTimeInEpochSeconds	uint64	Creation time of this message.
expirationTimeInEpochSecond	uint64	The expiration time of this message. After this time, the message shall be removed from the system

5.3 HazardMessage

HazardMessage component contains details about Hazard identifiers, times, location as well as the Hazard message.

Attribute	Data type	Description
id	Identifier	See 5.3.1
location	Location	See 5.3.2
times	Times	See 5.3.3
hazard	Hazard	See 5.4

5.3.1 Identifier



Attribute	Data type	Description
id	string	Unique identifier for a hazard message.
version	string	Version for Hazard message. For a new message version will be set as 1. For update messages, the version number will be incremented by 1.

5.3.2 Location

Location details of the Hazard message.

Attribute	Data type	Description
type	LocationType	The type of location.
openlr	OpenLR	The binary OpenLR location reference represented with multiple encodings.
segmentIds	SegmentIds	List of representations of the location using segment IDs of the underlying map.
coordinate	Coordinate	A single geo-coordinate in degrees according to the WGS-84 schema.
bearingInDegrees	uint32	Direction in degrees [0-360°] measured from North, clockwise.
frc	Frc	The dominant functional road class represents an order of importance for a road with respect to the overall road network. Different map data providers use different classifications

5.3.3 SegmentIds



Attribute	Data type	Description
type	ReferenceType	The type of IDs used to reference segments. OSM_WAY_ID is the only supported type.
segmentId	SegmentId	Sequence of SegmentIds representing the location using the specified ID type.

5.3.4 SegmentId

Attribute	Data type	Description
id	int64	Unique identifier of the segment
backwards	bool	True, if the segment is to be interpreted in the reverse direction
startOffsetInMeters	int32	The offset in meters from the start of the segment.
endOffsetInMeters	int32	The offset in meters from the end of the segment.

5.3.5 Times

Attribute	Data type	Description
reportingTimeInEpochSeconds	uint64	The time, when this message has been reported the first time.
startTimeInEpochSeconds	uint64	The expected start time of the hazard
endTimeInEpochSeconds	uint64	The expected stop time of the hazard.

5.4 Hazard

Main component for the hazards content. The hazard category and hazard type are mandatory attributes and define the nature of the hazard. The other attributes inform about the likelihood

that the hazard occurs, reporting current severity of the hazard, or finally if more detailed information is available.

Attribute	Data type	Description
category	HazardCategory	See 5.5
type	HazardType	See 5.6
likelihoodOfOccurrence	uint32	The likelihood that the hazards occur represented as percentage [0%...100%]
severity	HazardSeverity	See 5.7
description	LocalizedString	A language dependent description of the hazard. There may be several languages supported
lastUpdatedTimeInEpochSeconds	uint64	The time when this message has been updated the last time
detailInformation		See 5.8

5.5 HazardCategory

Defines the category of the hazard.

Category	Value
CATEGORY_UNSPECIFIED	0
CATEGORY_GENERIC	1
CATEGORY_TRAFFIC	2
CATEGORY_ROAD	3
CATEGORY_WEATHER	4

5.6 HazardType

Defines the type of the hazard.

Type	Value
TYPE_UNSPECIFIED	0
TYPE_JAM_TAIL_WARNING	1
TYPE_GENERIC	2
TYPE_ACCIDENT	3
TYPE_BROKEN_DOWN_VEHICLE	4
TYPE_SLIPPERY_ROAD	5



TYPE_REDUCED_VISIBILITY	6
TYPE_STRONG_WIND	7
TYPE_OBJECTS_ON_ROAD	8
TYPE_BAD_ROAD_CONDITIONS	9
TYPE_ROADWORKS	10
TYPE_EMERGENCY_VEHICLE_APPROACHIN	11
TYPE_WRONG_WAY_DRIVER	12

5.7 HazardSeverity

The severity of a hazard describes its criticality with respect to the danger.

Severity	Value
SEVERITY_UNSPECIFIED	0
SEVERITY_CRITICAL	1
SEVERITY_MAJOR	2
SEVERITY_MEDIUM	3
SEVERITY_LOW	4

5.8 DetailInformation

Specific properties of each type of hazard.

5.8.1 BadRoadConditionsDetailedInformation

Attribute	Data type	Description
isPothole	bool	True indicates that potholes cause the bad road conditions.

5.8.2 JamTailWarningDetailedInformation

Attribute	Data type	Description
speedAtTailInKilometersPerHours	int32	Speed at the end of the queue. A significant drop in speed from normal driving to the end of the queue is expected.

5.8.3 EmergencyVehicleApproachingDetailedInformation



Attribute	Data type	Description
speedInKilometersPerHour	int32	Speed of the Emergency Vehicle in Kilometers per hour.

5.8.4 ObjectsOnRoadDetailedInformation

Attribute	Data type	Description
animal	bool	True, if the obstacle is one or more animals. False, if the obstacle is not an animal. Not available means that the service does not know if it is an animal or not.
people	bool	True, if the obstacle are people. False, if the obstacle are not an people. Not available means that the service does not know if there are people or not.

5.8.5 ReducedVisibilityDetailedInformation

Attribute	Data type	Description
dueToFog	bool	True, if the main reason for reduced visibility is due to fog. False, if the main reason for reduced visibility is not due to fog. Not available, if the main reason is not known or related to a different cause.
dueToHeavyRain	bool	True, if the main reason for reduced visibility is due to heavy rain. False, if the main reason for reduced visibility is not due to heavy rain. Not available, if the main reason

		is not known or related to a different cause.

5.8.6 RoadworksDetailedInformation

Attribute	Data type	Description
lengthAffectedInMeters	int32	The length of the roadworks in meter.
currentSpeedInKilometersPerHour	uint32	The current average speed in kmh for the roadworks stretch.
laneChangesExpected	bool	Lane changes can be a reduced number of lanes, narrow lanes or a different lane structure throughout the roadworks stretch. True, if lane changes are expected. False, if no lane changes are expected. Not available, if it is not known if the roadworks hazard also includes lane changes.

5.8.7 StrongWindDetailedInformation

Attribute	Data type	Description
forHighSidedVehicles	bool	True, the strong wind hazard affects mainly high sided vehicles. False, the strong wind does affect all vehicle types. Not available means that a differentiation between high sided vehicles and others cannot be made.

6 Appendix

6.1 Message example



The example below shows the textual representation of a hazard warning event message. At runtime, Protocol Buffers will be delivered in binary form.

```
metaData {
 expirationTimeInEpochSeconds: 1681989834
}
message {
 id {
  id: "2692db19-a4b1-38a2-8595-1d453b6fc5b3"
  version: "1"
 }
 location {
  type: LINEAR
  openlr {
   base64: "C7LEfxvPFzPYDfxh//4zKOc="
  }
  coordinate {
   longitudeInDegrees: -108.6081
   latitudeInDegrees: 39.10631
  }
  bearingInDegrees: 122
  frc: FRC_6
 }
 times {
  reportingTimeInEpochSeconds: 1681918920
 }
 hazard {
  category: CATEGORY_ROAD
  type: TYPE_ROADWORKS
  likelihoodOfOccurrence: 30
  severity: SEVERITY_LOW
  roadworksDetailInformation {
   lengthAffectedInMeters: 76
  }
 }
}
```